

AI Applications on Humanoid Robots

Group: 21
ID's: 2230765031
2230765040
Course Name: AIN101
Instructor Name: Suat OZDEMIR
Submission Date: 31.12.2024

Abstract

Humanoid robots are advanced machines designed to resemble and imitate the physical structure, behavior and functions of the human body. These robots often have human-like features such as a head, torso, arms and legs, allowing them to perform tasks in environments created for humans. Humanoid robots aim to bridge the gap between humans and machines using human-centered design and interaction models. Humanoid robots develop their consciousness by communicating with humans and begin to think like humans over time. Technologies used in producing these humanoid robots integrated into humanoid robots include advanced robotic hardware, artificial intelligence (AI), computer vision, natural language processing (NLP) and machine learning. These allow humanoids to perceive their environment, make decisions, communicate effectively, and perform complex physical actions. Our aim in this report is to explain why people want to make humanoid robots, to explain these features of robots, to address them from an economic and ethical perspective, to briefly explain their stories and to explain how they will be shaped in the future.

Contents

1. A Brief History of Robotics (2-3)
2. Examples of Humanoid Robots (3-4)
3. Why Humanoid Robots Be Produced (4-6)
4. Ai technologies that used in Humanoid Robots
 - 4.1. Seeing and Perceiving the Outside World
 - 4.1.1. Understanding (7-8)
 - 4.1.2. Understanding Shapes and Objects (8-9)
 - 4.1.3. Pathfinding Algorithms (10-11)
 - 4.2. Sound Processing and Using (11-12)
 - 4.2.1 Natural Language Processing (NLP) and Emotions (12-14)
 - 4.3. Touch Perception (14-16)
 - 4.4. Effective Movement (16)
 - 4.4.1. Learning Reflexes (16-17)
 - 4.4.2. Training Movement in Virtual Reality (17-18)
5. Economic Aspects of Humanoid Robots (18-19)
6. Ethical Aspects of Humanoid Robots (19)
7. Future of Humanoid Robots (20-21)
8. Conclusion (21)

Introduction

The robotics industry is developing day by day. The biggest reason for this development is robots can be used in every area of life such as military bases, workplaces, homes and restaurants. But every area has some special requirements for these robots. For instance, a robot that works in a kitchen must be able to use an oven and be able to hold a pan. According to these requirements, robots are changing and creating several types of robots such as Industrial Robots, Home Robots, Medical Robots. One of the important types of robots is Humanoid Robots. The most important feature of humanoid robots is that, as the name suggests, they have a shape like human structure. They have a head, torso, arms and legs like us. Compared to other robots, humanoid robots are more advantageous because they can do every job that humans can do. For this reason, these days so many companies try to make humanoid robots in their field, and they develop recent technologies about ai or mechanics of humanoid robots. In these days understanding these technologies is important because Humanoid robots will take up more space in the robotics industry in the future. But if you want to understand how humanoid works, firstly we must understand what robotics and robots are and how they came to these day.

1. A Brief History of Robotics

The concept of human-like automatons is nothing new. Already in the second century B.C., the Hero of Alexander constructed statues that could be animated by water, air and steam pressure. In 1495 Leonardo da Vinci designed and possibly built a mechanical device that looked like an armored knight. It was designed to sit up, wave its arms, and move its head via a flexible neck while opening and closing its jaw. By the eighteenth century, elaborate mechanical dolls were able to write short phrases, play musical instruments, and perform other simple, life-like acts. In 1921 the word robot was coined by Karel Capek in its theatre play: R.U.R. (Rossum's Universal Robots). The mechanical servant in the play had a humanoid appearance. The first humanoid robot to appear in the movies was Maria in the film *Metropolis* (Fritz Lang, 1926). Westinghouse Electric Corporation exhibited at the 1939 and 1940 World's Fairs the tall motor man Elektro. Humanoids in appearance, it could drive on wheels in the feet, play recorded speech, smoke cigarettes, blow up balloons, and move its head and arms. Elektro was controlled by 48 electrical relays and could respond to voice commands. Humanoid robots were not only part of western culture. In 1952, Osamu Tezuka created Astroboy, the first and one of the world's most popular Japanese sci-fi robots. In 1973 the construction of a human-like robot was started at Waseda University in Tokyo. Wabot-1 was the first full-scale anthropomorphic robot able to walk on two legs. It could also communicate with a person in Japanese and was able to grip and transport objects with touch-sensitive hands. The group of Ichiro Kato also developed Wabot-2, which could read music and play an electronic organ. It was demonstrated at the Expo 1985 in Tsukuba, Japan. Wabot-2 was equipped with a hierarchical system of 80 microprocessors. Its wire-driven arms and legs had 50 degrees of freedom. Many researchers have also been inspired by the movie *Star Wars* (George Lucas, 1977) which featured the humanoid robot C3-PO and by the TV series *Star Trek - The Next Generation* (Gene Roddenberry, 1987) which featured the humanoid Data. In 1986 Honda began a robot research program with the goal that a robot "should coexist and cooperate with human beings, by doing what a person cannot do and by cultivating a new dimension in mobility to ultimately benefit society." After ten years of research, Honda introduced in 1996 P2 to the public, the first self-contained full-body humanoid. It was able to walk not only on flat floors but could also climb stairs. It was followed in 1997 by P3 and in 2002 by Asimo. In the U.S. Manny, a full-scale android body, was completed by the Pacific Northwest National Laboratory in 1989. Manny had 42 degrees of freedom, but no intelligence or autonomous mobility. Rodney Brooks and his team at MIT started in 1993 to

construct the humanoid upper-body Cog. It was designed and built to emulate human thought processes and experience the world as a human. Another milestone was the Sony Dream Robot, unveiled by Sony in the year 2000. The small humanoid robot, which was later called Qrio, was able to recognize faces, could express emotion through speech and body language, and could walk on flat as well as on irregular surfaces. More recent examples of humanoid robot appearances in the movies include David from A.I. (Steven Spielberg, 2001), and NS-5 from I, robot (Alex Proyas, 2004).

2. Examples of Humanoid Robots

Elektro and Sparko: The first robot (automaton) was presented at the world's fair in 1939.

The Rancho Arm: The first assistive robot was developed at a hospital in California and later controlled by a computer at Stanford University in 1963. The robot was composed of six joints used as a tool for the handicapped. This is the first robotic arm controlled by a computer.

Stanford Robotic Arm: The first electrically powered computer-controlled robot arm in 1969. The robot used a set of optical and contact sensors.

The Tentacle Arm: Marvin Minsky designed a tentacle arm robot inspired by an octopus in 1970. It had more than twelve tentacles that allowed the robot to go around obstacles.

The Silver Arm: MIT designed the first robot that is able to assemble small parts in 1974. The robot was equipped with pressure and touch sensors to mimic the soft touch of human fingers.

FRED Robot: The first playful robot was developed by Atari engineering in 1982. It never made it to the market.

Denning Sentry Robot: The first security sentry robot was developed in 1985, which could patrol for more than 14 hours and alert about anything unusual in a range of 150-foot radius.

Mitsubishi Movemaster: It was the first small robotic arm gripper which could perform tasks such as assembling small products or handling chemicals. It was presented in 1987.

Aibo Robot: First robotic pet dog. It could “learn”, interact with its environment and responded more than 100 voice commands. It was presented in 1999.

Asimo: Robot Asimo from Honda presented the first humanoid robot that could walk like humans, climb stairs, change its direction and detect hazards using a video camera in 2000.

Robotnaut 2: NASA revealed a humanoid robot with a wide range of sensors that can replace human astronauts in 2010.

iCub: A 1-meter-high humanoid robot for research in human cognition at IIT, Italy in 2010. The robot can express emotions and is equipped with tactile sensors to interact with the environment.

Pepper: Japanese company Softbank presented the first robot in 2014, so-named Pepper, to be used for customer service. The robot has integrated an emotion engine to interact with people.

Robot exoskeleton: A complete paralyzed man was able to walk again using a robotic exoskeleton designed by Ekso Bionics in 2014.



3. Why Humanoid Robots Be produced

The ability to recognize that others, like us, have the capacity to understand, infer, and attribute emotions, motivations, and goals is called theory of mind. The capacity to attribute a mind to others is considered central to human functioning because it opens up the possibility of a common ground for interaction.

Humans' preference for human-centered interactions is often presented as the underlying reason for humanizing robots, i.e., if humans mindlessly apply the rules of human-human

interaction to interactions with nonhumans and objects, humanizing robots will result in more natural and efficient HRIs. Some socio cognitive processes in the humanization of robots are:

❖ Inferring agency from biological motion

People can infer qualities and attribute intentions based on minimal information. Research on the recognition of biological motion shows that people can recognize gender, personality traits, emotions, and even complex actions such as dancing just by watching a film of an actor with lights attached to the main joints of the body while moving in a dark environment. As such, it is not surprising that research with robotic appliances like the Roomba, the vacuum cleaning robot, has found that users describing its random movements as gentle or clumsy and attributing the Roomba a sort of proto-personality.

❖ First impressions and the anthropomorphic form

First impressions based on physical appearance have been shown to play a significant role in social interactions leading to the attribution of such diverse traits as attractiveness, trustworthiness, competence, and aggressiveness, based only on a 100 ms exposure to a face. When designers attempt to humanize robots through physical cues, they should be aware that their creations will be subjected to these quick evaluations. Research has shown that people not only make inferences about a robot's intelligence, competence, warmth, or friendliness, based on design options like face traits , height , gender , or speech and facial expressions , but also make decisions about what jobs a robot should perform based on a robots' appearance . Since first impressions can influence expectations and motivations for interaction, anthropomorphic robots should be designed to reduce possible expectation gaps.

❖ Psychological account of anthropomorphism

Although psychological anthropomorphism will follow a developmental path along the life cycle, some individuals will show a greater tendency to attribute human motives to nonhuman animals and objects. Cultural differences in the definition of what it is to be a human, an animal or an inanimate object also play part in the process. Of specific interest for the design of social robots is the fact that the perceived similarity of the target (e.g., similar motion, similar morphology), induces people to base their reasoning more on egocentric knowledge.

❖ The computers are social actors paradigm

The computers are social actors (CASA) paradigm follows from the observation that people mindlessly apply social norms in their interaction with computers . That is, given an object with a sufficient set of cues (e.g., language, interactivity, a traditionally human role), people tend to mindlessly apply social rules (e.g., reciprocity and politeness), social categories (e.g., gender stereotypes), and infer personalities. The idea that a small set of cues can automatically and unconsciously elicit human–human interaction social scripts has been very influential, for both human–computer interaction and HRI research, and paved the way for the current interest in humanizing both virtual and physical robotic agents.

There are some benefits to producing humanoid robots. One of the most prominent positive effects of humanizing robots is to establish mutual psychological understanding within an interaction between humans and robots . This can be achieved by programming social robots with some key-ingredients in their design, such as the notion of “being there,” being human-aware, and being understood . Thus, robots that resemble humans may create familiarity in order to grounding interactions in already established human skills and social norms and may thus facilitate HRI. Several benefits have been reported regarding the programming of social robots that mimic intrinsic human abilities, many of which have been used in applied areas for healthcare and therapy, education and entertainment. In healthcare, the use of robots has been associated with several benefits such as decreasing anxiety in hospitalized children. In educational contexts, research has also shown that children can achieve relevant learning gains in school curricular topics when exposed to long-term interactions with robots that mimic empathic capabilities (such as contingency behaviors and personalization during learning tasks), in contrast with short-term interactions with robots not endowed with empathic capabilities. Additionally, children have shown to perceive robots that mimic empathy as their friends, even when explicitly instructed that the robot was behaving as their tutor, which indicates that interactions with robots capable of mimic human capabilities may lead to positive changes in the perception of a robot's role. In a study in which adults played a chess game with a robot programmed to mimic empathy (defined as the ability to produce an emotional/cognitive response upon the understanding of another's emotion), the robots mimicking empathy were perceived as friendlier than those that do not.

4.1.1. Understanding

Here we explain how robots learn visually. To be useful in everyday environments, humanoid robots must be able to adapt to existing abilities and cope with changes. They also need to learn new skills quickly. Fortunately, humanoid robots have the unique possibility of learning from skilled teachers, the people around them. This is called imitation learning or programming by demonstration. Imitation learning has been applied to complex movements and manipulation tasks, such as swinging a tennis racket or producing gestures. One challenge of imitation learning is the perception of the teacher. Often, external motion capture systems that rely on special markers or attached motion sensors to detect the teacher's movement are used. Another challenge is the mapping between the human body and the robot's body. Some human movements may not be possible for the robot, for example, due to lack of joints, limited joint angles, or dynamic constraints. On the other hand, the robot may have degrees of freedom that are not constrained by the captured motion. These must be controlled by optimizing secondary criteria such as energy usage. A frequently used possibility to simplify imitation is to rely on kinesthetic teaching, where the teacher directly moves the robot's limbs. Although imitation of human movement patterns greatly helps in the production of human-like movement, it is not successful. Because of the differences between the teacher and the robot, true imitation requires the robot to infer the human teacher's intentions and develop its own strategy to achieve the same goals.

Another possibility for optimizing the behavior of human-like robots is reinforcement learning. Here, the robot simulates rewards or punishments as it interacts with the environment. There are many techniques for reinforcement learning. Among the most promising techniques are stochastic gradient techniques used to optimize bipedal walking patterns and natural actor-critic learning, which has been shown to learn to hit a ball with a stick. One of the general difficulties with reinforcement learning is the generation of rewards. It cannot be assumed that the environment produces a sufficient reward structure for learning complex tasks. On the contrary, intermediate rewards should be generated by the robot itself when successfully performing subtasks or satisfying internal needs. On the other hand, the robot should generate negative rewards when it violates internal constraints, for example, when falling. Learning from several examples is especially important in such cases. In addition to learning behavior, learning and adaptation are also important for perception.

For example, computer vision and speech recognition must adapt to changing light conditions and different auditory backgrounds. Robots need to get to know new objects in their environment and also, learn new words and names. For navigation, some humanoid robots create maps of their environment. An example of successful mapping with a humanoid robot was presented by Gutmann et al. using the Sony Qrio.

4.1.2. Understanding Shapes and Objects

We demonstrate the image processing mechanism in humanoid robots with a robot made by Japan called HRP-2P. HRP-2P has a stereo vision system composed of three cameras (figure 1). Two horizontal cameras are separated by 120 mm, and the third camera is 60mm upper than them. three cameras system be adopted since it is difficult for stereo vision composed of two cameras to detect the horizontal line. The weight of the vision system is less than 700 g. Relatively short focus lens is used in order to measure a position of an object, whose standoff is from 0.5m to 4 m. The camera's shutter speed is controllable by a computer to adapt various lightning conditions. The image processing is based on VVV(Versatile 3D Vision System) System.

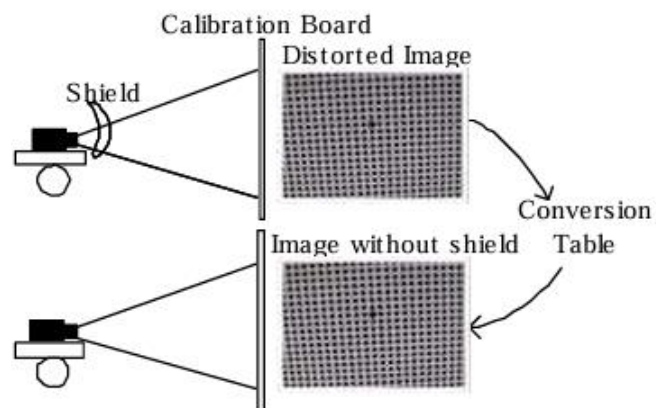


(Figure 1)

VVV consists of several image processing modules and it can recreate a 3D shape model, detect an object, measure its position and track an object with it.

By means of a plastic shield in front of the cameras, the object image is distorted. It is practically difficult to model the shield shape

and the position of the camera accurately. Therefore, a transformation table is made between a distorted image and an image without a shield. On a calibration board, the circle patterns are arranged at equal intervals and a larger circle is arranged near the center of the board. The board is captured with and



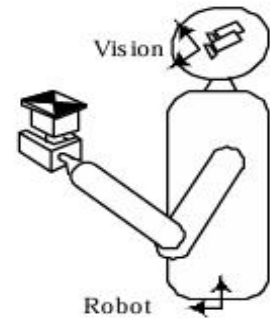
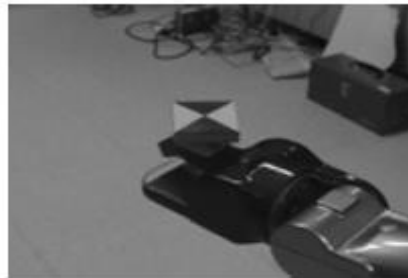
(Figure 2)

without a shield (Figure. 2) and the position of each circle is determined.

The larger circle position in each image is used as the positioning reference of all other circles and the correspondence of all circle points between the two images is obtained. Using an interpolation of this correspondence, a transformation table is made for each pixel. Also, the lens distortion is corrected using the same image without a shield. As a result, the distortion is reduced to 0.2 pixels.

The perceived location of an object is represented by an image coordinate system. To use this location data, the vision coordinate system must be transformed into the robot coordinate system. A marker is placed on the robot finger and the HRP-2P moves the hand to several locations in the field of view.

The HRP-2P captures the image and obtains the location of the finger at each location. Us-



(Figure 3)

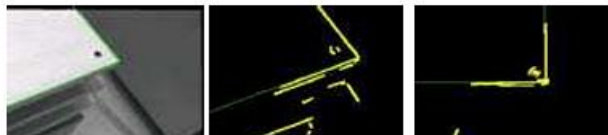
ing the detected marker locations in the vision coordinate system and the finger locations in the robot coordinate system, the transformation matrix from the image coordinate system to the robot coordinate system is calculated.



a) Detecting Panel



b) Captured Images



(Figure 4)

For HRP-2P to grasp and move a panel with a human operator, an object recognition function is required. A model-based recognition system is used, and the grasping point is calculated from the results of recognizing the top edge of the panel with an error of 2 mm. Figure 4 shows (a) the HRP-2P detection panel, (b) the captured images, and (c) the result of matching a panel model.

4.1.2. Pathfinding Algorithms

In robotics, robot path making is a fundamental problem. An efficient robot needs to locate the ideal way which decreases the number of steps to be taken between the starting point and the target point. The most efficient method found for this today is Using Neural Network (NN) methods. Using Neural Network (NN) methods a humanoid can prepare a path for the robot through which robot can not only reach the target but also avoid the static and dynamic objects. The study that shows us that this is the most efficient method is the pathfinding algorithm on ROBONOVA-1.

ROBONOVA-1 is a humanoid robot consisting of 24 servo motors. .It has got 24 DOF. It has got sensors like a gyro sensor, acceleration sensor. The servo motors are used for the accurate positioning of various links of the ROBONOVA humanoid. The ROBONOVA equipped with a remote control trans meter and receiver. The ROBONOVA has got lithium polymer battery for getting power. The battery can be used for several hour and this battery can be exchanged during navigation and can be used fruitfully. In that study the strategy depends on the utilization of Laplace's Equation to compel the potential function generation. The final output from the neural network can be derived by using the equations (1) and (2).

$$\text{Steering angle} = f[(\text{summetion of } m \text{ INPUTS})^4] \quad (1)$$

$$(\text{summetion of } m \text{ INPUTS})^4 = \sum_l \text{WEIGHT}_{lk}^4 \cdot \text{INPUT}_i^3 \quad (2)$$

Front obstacle distance [FOD], left obstacle distance [LOD], right obstacle distance [ROD], heading angle [HA] are chosen as the inputs to the controller. The inputs are processed by two levels of hidden layers and the output of the controller is obtained in the form of steering angle. To test the working of the Neural Network controller multiple simulations are performed. A specific arena was selected and obstacles were positioned at random locations. The robot starts its journey from the predefined initial position and reaches the destination by the help of the controller. The experimental results are recorded in six steps. They are represented in Figs 5a-5f. Fig 5a shoes the starting condition, and Fig 5f Shows the final position and path covered during navigation in simulation mode.



(Figure 5a)



(Figure 5b)



(Figure 5c)



(Figure 5d)



(Figure 5e)



(Figure 5f)

4.2. Sound Processing and Using

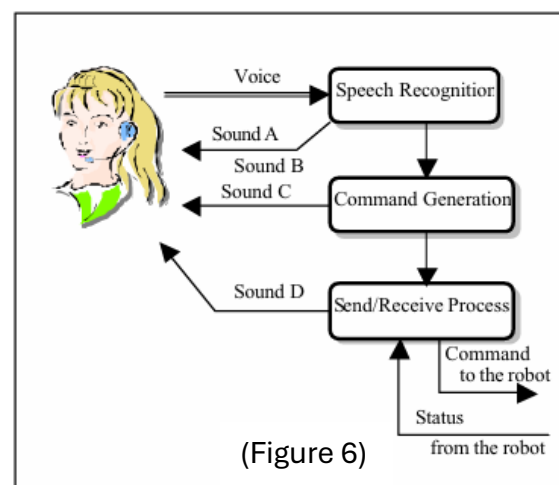
We demonstrate the sound processing mechanism in humanoid robots with a robot made by Japan called HRP-2P. Voice instructions were developed and installed on a laptop carried by a human. Communication occurs between the PC and the robot. The PC recognizes the speech, converts it into a command and transmits it to the robot. After interpreting it, the

robot executes the command. The human also receives a confirmation message from the PC through the headset that the command has been sent to the robot. The human also sees the robot if he executes his command. However, the problem occurs when a command is not sent to the robot due to the following failures.

- 1) Failure in speech recognition
- 2) Failure in communication

Also, the whole practical system, recognition and communication takes some time. Sometimes this time frame can create a new problem. For example, if the time is long and the human is restless, the human may send the same command twice before the robot executes the first command. In this case, the robot will repeat the same action during the execution of the second command. All these problems can be solved by developing a system that sends information in various stages as shown in the flow chart in Figure 6. The system provides information to humans using different sounds in each stage as shown below.

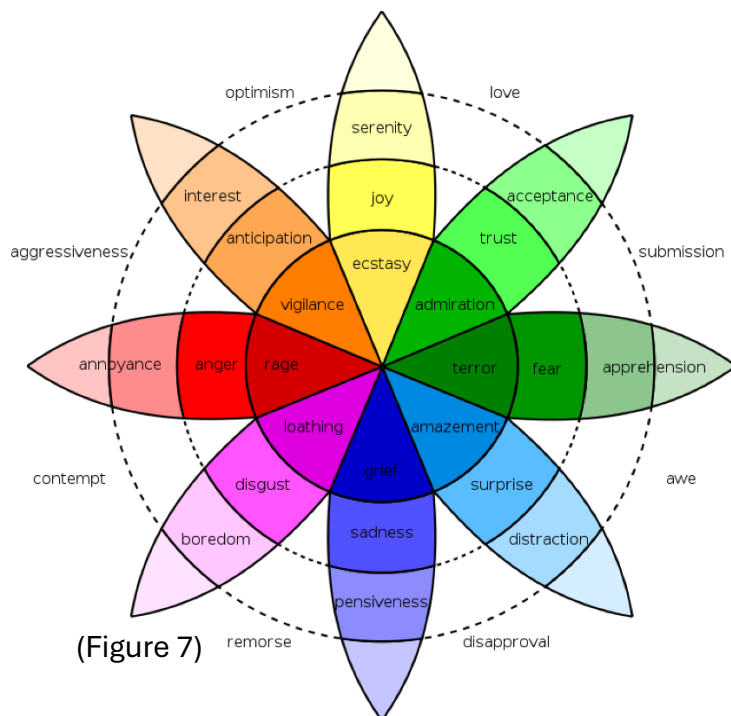
- 1) Speech recognition is started (Voice A)
- 2) Recognition is impossible (Voice B)
- 3) Outside of commands (Voice C)
- 4) Communicating (Voice D) With this development, it is possible for humans to know the status of the system without using special devices and to transmit commands to the robot with certainty.



4.2.1 Natural Language Processing (NLP) and Emotions

For emotion recognition systems, an important aspect to consider is the emotion models that delimit the classification process. While according to Cowie, in the field of psychology, there are many different theories regarding the representation of emotions, within the area of NLP, there are two that stand out as the most used: Ekman's basic emotions and Plutchik's wheel of emotions. Ekman's model consists of six basic emotions: anger, disgust, fear, happiness, sadness, and surprise. Plutchik's model consists of a multi-dimensional

approach, where there are four opposing pairs of axes and emotions are defined as different points along these axes. In this model, an emotion is determined by an emotion axis and its intensity. These axis pairs are joy–sadness, anger–fear, trust–disgust, and surprise–anticipation. Figure 7 shows an extraction of the Plutchik’s model, with these axis and intensity denoted with colors in the concentric circles; from these emotions, many other emotions can be derived as a combination of other emotions and intensities. Most works in emotion detection consider a small subset of these group of emotions. Most common approaches to performing emotion classification and general NLP tasks are based on recurrent neural networks (RNN) or some variants of them, such as long short-term memory (LSTM, multiplicative LSTM (mLSTM) or gated recurrent unit (GRU).



However, these approaches suffer from a couple of related issues: the complex nature of emotion expression and the shortage of quality data for this task. Finding quality label data for this task is hard to do, at least in order to completely train these types of models. Nevertheless, these limitations have been overcome by an attention-based transformers architecture, which is inspired by the encoder–decoder architecture used for se-

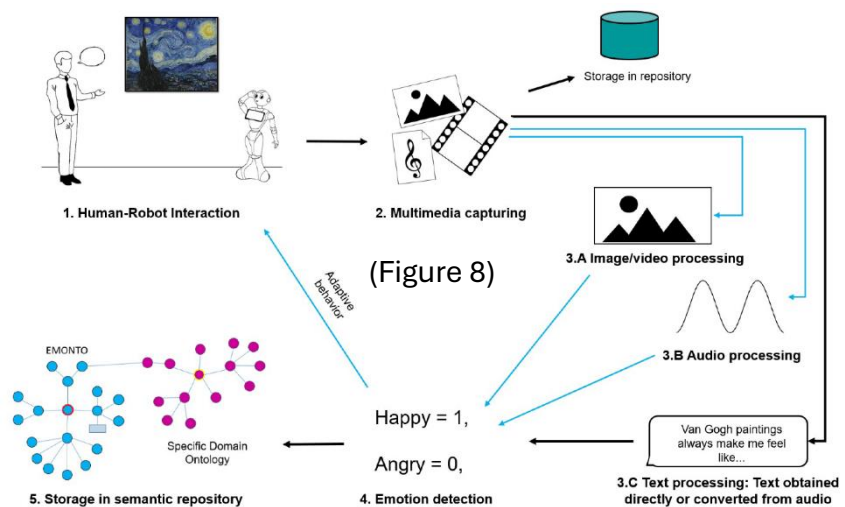
quence to sequence (Seq2seq) tasks. In high-level terms, it consists of two stacks (one of encoders and one of decoders), where the encoder and decoder use some form of self-attention and a feed forward neural network. Each encoder in the stack receives its input from the previous layer, except the first one that receives a series of tokens. Then, the final encoder in the stack sends its output to every decoder. Then, each decoder uses this and the input from the previous layer in the stack to return its output. Afterward, the output for the final decoder is passed to a linear + SoftMax function to finally return the output of the network. Recently, it has become more and more common to use transformers for NLP in texts, since they have been proven to be effective in many different tasks (e.g., masked token prediction, next

sentence prediction, question answering, machine translation, summarization, natural language inference, and sentiment analysis). This is thanks to the fact that they are trained with unsupervised learning models, diminishing the need for labeled data. Transformers are very effective in transfer learning, allowing researchers to pretrain these transformers with large amounts of general-purpose texts and then to finetune these models for their specific tasks with good results, less effort, and labeled data, such as NVIDIA-AI (currently named Megatron-LM), the model is proposed by researchers of NVIDIA.

The proposed framework is a process divided into five sequential steps, as shown in Figure 7: (1) *human–robot interaction*; (2) *multimedia capturing*; (3) *multimedia processing: image/video processing, audio processing, and text processing* (text obtained directly or converted from audio); (4) *emotion detection*; and (5) *storage in a semantic repository*.

From the human–robot interaction (step 1 in Figure 8), multimedia content is gathered from the robot’s sensory capacity; these raw data are saved in a repository (step 2 in Figure 8). The following steps imply transformations of the obtained data from the previous step, starting with the conversion of raw

input obtained from the user to proper representations for emotion analysis (e.g., feature vectors and text) (step 3 and step 4 in Figure 8) and finishing with the population of an emotion ontology that can be used for different purposes (step 5 in Figure 8).



This framework can be instantiated in social robots systems, taking advantage of all the sensing capacities of robots.

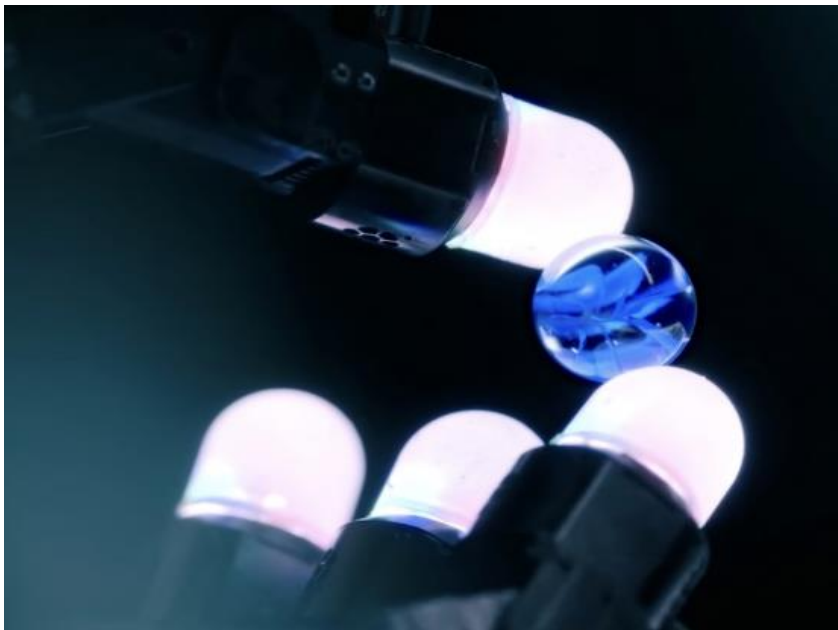
4.3. Touch Perception

Of all our senses, touch arguably is the most critical in how we interact with the world and how we explore the world. It enables us to measure forces and recognize object properties – shape, weight, density, textures, friction, and elasticity. It also plays a significant role

both in social relationships and in cognitive development. Experts in the field of Humanoid Robot are trying to mimic the familiar human characteristics towards the advancement of robotic hand manipulation. Hand: Fingers. They hypothesize that the next generation of touch digitization in robotics will enable intelligent applications. Systems for discerning significantly higher levels of physical information during environmental interaction.

One of them that is trying to mimic the familiar human characteristics towards the advancement of robotic hand manipulation is Meta Company. Meta is a company that has been dedicating itself to recent technologies for a long time. Some of these recent technologies are Sparsh, Digit360 and D.

Digitizing touch depends on two fundamental features: one temporal in nature and one spatial. Temporal features process information from signals of time variation, whereas spatial features process a discrete multidimensional array of geometrical signals. Meta combine these methods within a unified platform to significantly improve the capabilities of touch digitization: a modular, finger-shaped, multi-modal tactile sensor with on-device artificial intelligence capabilities and superhuman performance. Digit 360.



(Figure 9)

To effectively capture the nuances in touch interactions with the world, an artificial fingertip must be sensitive in both temporal and spatial domains. These domains are obtained as modal signals encompassed within the artificial fingertip through visual, audio, vibration, pressure, heat, and gas sensing.

In DIGIT 360, Meta encompass the touch digitization system into the form of an artificial fingertip with a similar geometry to a human finger. The surface of the fingertip encodes touch information from depressions in a reflective layer of which internal light

reflections are captured by the camera. Digit 360 is designed to advance spatial, temporal, and multi-modal performance through the embodiment of a modular platform in touch digitization research.

To achieve the high spatial and temporal performance demonstrated by our artificial fingertip, we use Sparsh, a family of SSL (self-supervised learning) models that are able to support haptic sensors of various configurations, reducing the need for custom labels through pre-training on over 460k haptic images, and self-distillation in pixel and latent spaces.

In evaluations, we find that SSL pre-training for touch representation outperforms task and sensor-specific end-to-end training by 95.1% on average over, and Sparsh (DINO) and Sparsh (IJEPA) are the most competitive, indicating the merits of learning in latent space for tactile images.

4.4. Effective Movement

Another important thing about humanoid robots working mechanisms is that movement. An average person might walk or run approximately 100,000 to 200,000 miles in their lifetime, depending on their daily activity levels. So, moving somewhere to somewhere is a really important thing for us. Like healthy humans, also functional humanoid robot must be able to change positions. The difference between humans and humanoid robots is while humans use their brains and spinal cord to move, robots use algorithms and artificial intelligence.

4.4.1. Showing Reflexes

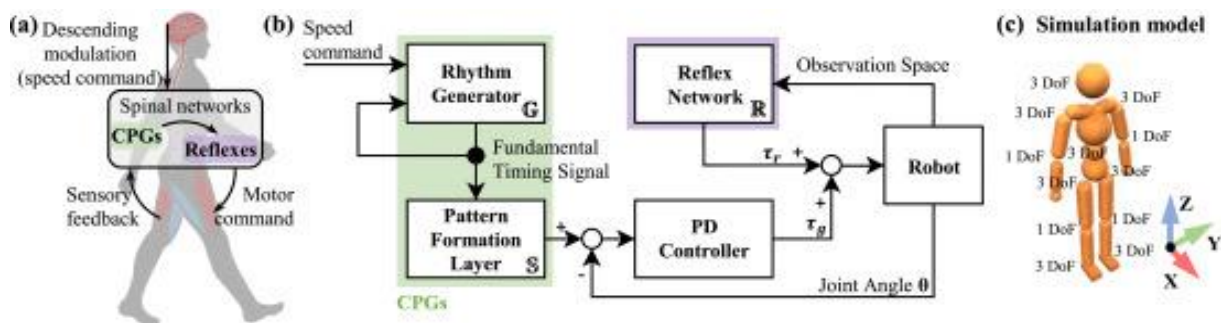
Despite the potential advantages of humanoid robots with bipedal locomotion, adaptively controlling humanoid robots is challenging because of their complex dynamics and overly redundant degrees of freedom (DoFs). One of the current issues that should be addressed is related to the gait coordination of humanoid robots, which can significantly affect their overall performance.

Inspired by neuroscience, central pattern generators (CPGs) are another promising approach for improving legged robots' locomotion. CPGs are neural circuits located in the spinal cord that generate rhythmic patterns of muscle activity, such as those used during walking and running. Using CPGs, robots can achieve more natural and stable movements, similar to those of

living organisms. The CPG mechanism involves a network of interconnected neurons that generate oscillatory signals that are transmitted to the muscles responsible for movement.

However, the question of how to effectively integrate and apply them to humanoid locomotion control remains unresolved, since CPGs can potentially constrain the control space and help decrease dimensionality but to be adaptive and flexible for different environments, it should be well supported by reflex networks.

In a study, one of the best ways to solve these problems is enhancing learning-based algorithms for humanoid locomotion using CPGs with a sensory feedback mechanism. In study they trained a CPG controller using imitation learning and then trained a reflex neural network using DRL. Unlike other algorithms that use reinforcement learning for imitation purposes, they training objective was not only to make the agent behave similarly to the collected human motion data; they used imitation learning to train a CPG-like controller to form feedforward control. Control framework, adaptive imitated CPG (AI-CPG), consists of rhythm generator G , which defines the rhythm of motor activities; pattern formation layer S , which shapes the rhythmic timing signals to the target joint angles of the robot; PD controller, which outputs the motor commands based on the error between current joint angles and target joint angles; and reflex neural network controller R based on sensory feedback.



(Figure 10)

4.4.2. Training Movement in Virtual Reality

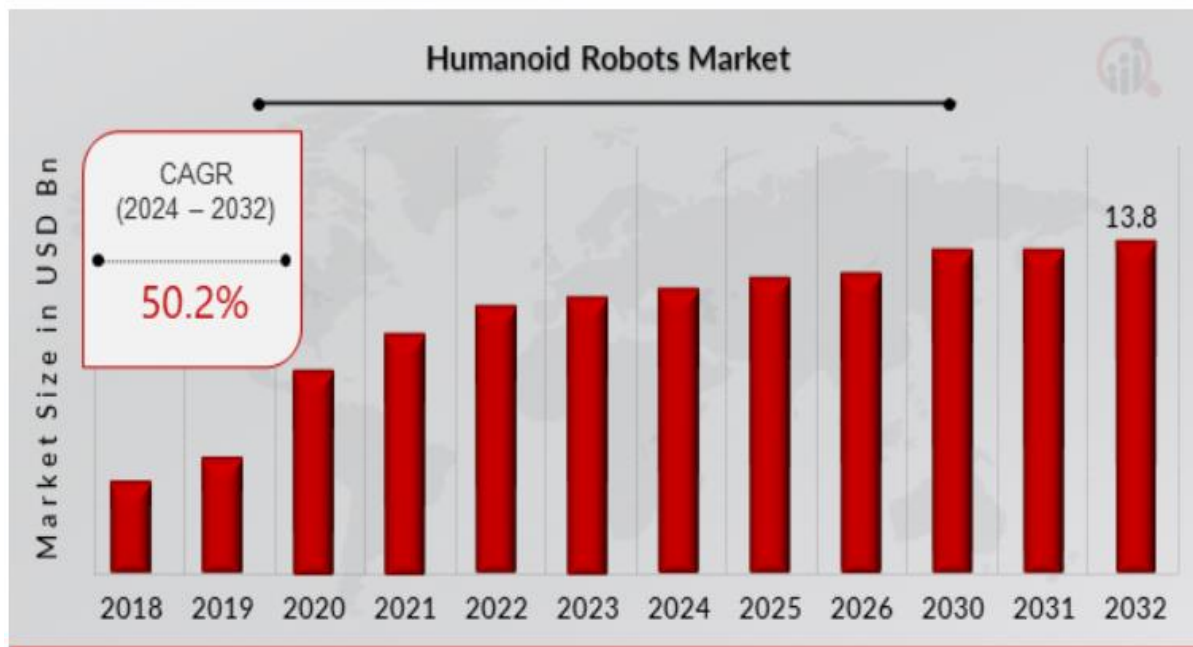
Training Humanoid robots' movement is a very costly process in terms of both space, time and finance. This is due to there is so many possible event or possible position for a humanoid robot that have two leg and two arm. A self-learning humanoid robot generally requires large facilities to accommodate its reinforcement learning process which also requires a team to supervise the entire process to ensure human and robot coexistence is harmonious.

Therefore so many people try to find cheapest methods for training movement of robots. One of the new founded method is using virtual environment for the trainings.

The proposed solution aims to replace this manual process with a virtual environment. This training method makes the robot believe that it is in a situation that is not actually there and behaves that way. To make environment physics more reactive to weight and size, quantitative data is vital to help design the length a robot can reach or its height, the size of the step it makes, and the weight of the robot. All of these quantitative data are required to assist develop this virtual world, including the robot's grip and how strongly or loosely it will grab the objects. Data is required to move further with the process of virtual environment design for robots to interact with objects virtually.

For creating virtual environment we need some different types of programs like; 3D rendering programs(3ds max, blender...), virtual physics motors(UE5...) and Virtual reality apps(Oculus...). Firstly, 3D elements created using 3D Max and later incorporated with UE5 to create and mesmeric environment for robots. Further, it offers a wide range of tools including lumen, and 3D rendering essential for lighting crucial to provide photorealistic simulation for different training scenarios.

5.Economic Aspects of Humanoid Robots



(Figure 11)

The global humanoid robots' market is estimated to reach a valuation of USD 13.8 billion by the year 2032 at a CAGR of 50.2% during the forecast period (2024-2032).It is

shown in figure 11. Humanoid robots are designed with human characteristics such as behavior, appearance, and body structure.

The developers in the humanoid robots market are working on solving issues such as dexterous manipulation, targeted application, bipedal locomotion, etc., to work in the same environment as humans. They can be useful for performing crucial tasks such as the dismantling of nuclear power plants, disaster emergencies, extra-vehicular activities, space exploration, etc. Thus, the development of machine learning, artificial intelligence, machine vision, etc., is a major driver of the Humanoid Robots Market. The market has various benefits, such as dealing with the distribution process and customer care. They are also durable, so they can visit challenging places for humans. It is estimated that the humanoid robots market will boom in the Asia-Pacific region in the upcoming years. Countries such as Japan and China are extensively focusing on the development of humanoids that will be favorable for the market.

6. Ethical Aspects of Humanoid Robots

The application of robot ethics is strongly related to the way robots are perceived. Acknowledging robots strictly as machines implies that they do not possess any hierarchically higher characteristics, nor should they be provided with consciousness, free will, or a level of autonomy superior to what is technically embedded by the designer. Yet, people's interaction experiences with robots fundamentally differ from people's experiences interacting with most other technologies due to a strong social or emotional in human-robot interactions. For instance, people act upon or talk to some robots as if the robot has sensations and emotions, even though existing robots are not sentient and lack feelings. Indeed, people's social responses to robots have been reported in many human-robot interaction studies. A lot of robot ethics discussions focus on the technological abilities of robots, i.e., what is in the 'mind' of the robot. It has been argued that the emphasis of robot ethics should shift to the robot's outside, i.e., what robots do to us. Although most people would reasonably agree that robots are programmed machines that only simulate social behavior, the same people seem to 'forget' this while interacting with these sociable machines.

7.Future of Humanoid Robots

Nowadays, with new AI technologies, the fact that there has been a great acceleration in the development of humanoid robots pushes many people to think. These include outrageous sci-fi stories, such as robots taking over the world, as well as very possible scenarios such as many people taking over their jobs. This makes it hard to assess the true functional positioning of robots, how this is likely to move forward and whether the outcome of progress could be detrimental to human society. But there is a fact that humanoid robots should advance into the future because they make it easier for us to understand many things about both humans and society; and the future of humanoid robotics depends on overcoming numerous challenges and leveraging emerging trends to advance the field.

Some of these **challenges** are;

1. **Biological Mechanism & Control:** Limited understanding of human anatomy and brain-like intelligence hampers progress in replicating human-like cognitive and motion control.
2. **Energy Efficiency:** Current energy sources and consumption rates are insufficient, necessitating innovations in biological energy conversion and efficient power systems.
3. **Technological Complexity:** Achieving natural movement and human-like interactions requires advanced AI, sensors, and mechatronics, alongside cost-effective production methods.
4. **Application-Specific Challenges:** Issues like high costs, legal/ethical concerns, user acceptance, and fault tolerance must be addressed to ensure broader adoption.

Some of these **future trends** are;

1. **Advanced Sensing and Perception:** Highly accurate, affordable sensors will enhance robot-environment interactions, enabling adaptive navigation and manipulation in complex environments.
2. **Enhanced Actuators:** New actuators, including artificial muscles and hybrid designs, will improve movement and load capacity.
3. **Interdisciplinary Research:** Collaboration across robotics, neuroscience, and material sciences will drive innovation in control algorithms, materials, and energy systems.
4. **Human-Robot Interaction:** Advances in multimodal communication, including speech and gesture recognition, will enable seamless human-robot collaboration.

5. **Energy Innovations:** Research into compact, high-capacity energy sources and efficient management systems will extend robot operating times.

Conclusion

Humanoid robots are a testament to the rapid advancements in robotics and artificial intelligence, showcasing their potential to reshape industries and daily life. By leveraging cutting-edge technologies such as AI, natural language processing, computer vision, and advanced movement algorithms, these robots are becoming increasingly adept at performing tasks, interacting with humans, and adapting to dynamic environments.

Despite these advancements, their development brings with it critical challenges and opportunities. On the economic front, humanoid robots offer enhanced efficiency and safety, particularly in high-risk scenarios, but they also pose the risk of job displacement in some industries. Ethically, the human-like features and behaviors of these robots demand careful consideration to ensure that their integration into society respects human values and fosters trust.

The future of humanoid robots depends on a balanced approach that aligns technological progress with human needs. Preparing the workforce through education and skill development, alongside establishing ethical frameworks, will ensure that these robots are used as tools to complement human effort rather than replace it. By addressing these challenges thoughtfully, humanoid robots have the potential to become reliable partners, enhancing human capabilities and contributing positively to society.

REFERENCES

- Kazuhiko YOKOYAMA, Hiroyuki HANDA, Takakatsu ISOZUMI, Yutaro FUKASE, Kenji KANEKO, Fumio KANEHIRO, Yoshihiro KAWAI, Fumiaki TOMITA and Hirohisa HIRUKAWA, International Conference on Robotics & Automation, Proceedings of the 2003 IEEE, pp2287-2289,2003
- , Joe Denny, Mohamed Elyas, Shannon Angel D’costa, Royson Donate D’Souza. Humanoid Robots – Past, Present and the Future, European Journal of Advances in Engineering and Technology, 2016, 3(5): 8-15
- Obaid Fareed, M. Irfan Anis. A hyper-realistic virtual environment for robots training, Springer-Verlag GmbH Germany, part of Springer Nature 2024
- Javier Andreu Perez, Fani Deligianni, Daniele Ravi and Guang-Zhong Yang, Artificial Intelligence and Robotics, UK-RAS Network, pp 25-26, 2024
- Mária VIRČÍKOVÁ, Peter SINČÁK. Artificial Intelligence in Humanoid Systems, Dept. of Cybernetics and Artificial Intelligence, FEI TU of Kosice, Slovak Republic, 2010
- Biswajit Sahoo, Dayal R. Parhi , Priyadarshi Biplab Kumar. Analysis of Path Planning of Humanoid Robots using Neural Network Methods and Study of Possible use of Other AI Techniques, Emerging Trends in Engineering, Science and Manufacturing (ETESM-2018)
- Humanoid Robot Market by Component (Hardware, Software), Motion Type (Biped, Wheel Drive), Application (Education and Entertainment, Research and Space Exploration, Personal Assistance and Caregiving) and Region – 2028 2023
- Maartje M. A. de Graaf, An Ethical Evaluation of Human–Robot Relationships, Int J of Soc Robotics , Springer , pp-291 292, 8:589–598 DOI 10.1007/s12369-016-0368-5, 2016
- Sven Behnke, Humanoid Robots – From Fiction to Reality? KI-Zeitschrift, 4/08, pp. 5-9, December 2008
- Carolina Higuera, Akash Sharma, Chaithanya Krishna Bodduluri, Taosha Fan, Patrick Lancaster, Mrinal Kalakrishnan, Michael Kaess, Byron Boots, Mike Lambeta, Tingfan, Mustafa Mukadam. Sparsh: Self-supervised touch representations for vision-based tactile sensing, FAIR at Meta, University of Washington, Carnegie Mellon University, Equal contribution
- Mike Lambeta, Tingfan Wu, Ali Sengül, Victoria Rose Most, Nolan Black, Kevin Sawyer, Romeo Mercado, Haozhi Qi. Digitizing Touch with an Artificial Multimodal Fingertip, FAIR at Meta, Learning, Adaptive Systems, and Robotics (LASR) Lab, TU Dresden, The Centre for Tactile Internet with Human-in-the-Loop (CeTI), University of California, Berkeley, October 31, 2024
- Graterol, W.; Díaz-Amado, J.; Cardinale, Y.; Dongo, I.; Lopes-Silva, E.; Santos-Libarino, C. Emotion Detection for Social Robots Based on NLP Transformers and an Emotion Ontology, MDPI, pp 5-8, 2021.
- Jean-Christophe Giger, Nuno Piçarra, Patricia Alves-Oliveira, Raquel Oliveira, Patricia Arriaga, Features of Emerging Technologies, Wiley Periodicals. Inc. pp 112-115, 2019
- Yuchuang Tong, Member, IEEE, Haotian Liu , and Zhengtao Zhang , Member, IEEE, CAA JOURNAL OF AUTOMATICA SINICA, IEEE, pp-307-317, 2024
- Scott Kuindersma, Robin Deits, Maurice Fallon, Andrés Valenzuela, Hongkai Dai, Frank Permenter, Twan Koolen, Pat Marion & Russ Tedrake. Optimization-based locomotion planning, estimation, and control design for the atlas humanoid robot, 31 July 2015, Volume 40, pages 429–455
- Shweta Singh; Devansh Chaudhary, Amar Deep Gupta; Bhanu Prakash Lohani; Pradeep Kumar Kushwaha; Vimal Bibhu. Artificial Intelligence, Cognitive Robotics and Nature of Consciousness, 3rd International Conference on Intelligent Engineering and Management (ICIEM), 2022, IEEE
- Guanda Li; Auke Ijspeert; Mitsuhiro Hayashibe. AI-CPG: Adaptive Imitated Central Pattern Generators for Bipedal Locomotion Learned Through Reinforced Neural Networks, IEEE Robotics and Automation Letters, (Volume: 9, Issue: 6, June 2024), pp- 5190 - 5197
- Gregory Kahn; Pieter Abbeel; Sergey Levine. BADGR: An Autonomous Self-Supervised Learning-Based Navigation System, IEEE Robotics and Automation Letters (Volume: 6, Issue: 2, April 2021), pp- 1312 - 1319